

SECTION C

10 **(short)** A metal sphere of radius 20 mm is placed in air with its centre 100 mm vertically above an infinite, horizontal, grounded, conducting plane and the sphere is then charged to 60 V relative to the plane. What is the approximate direction and magnitude of the electric field just above the conducting plane directly under the centre of the sphere? [10]

11 **(short)** The cross-section of a coaxial cable, which may be taken to be infinitely long, is shown in Fig. 8. It consists of an inner conductor of diameter 3 mm, and an outer conductor which has an inner diameter of 5 mm and an outer diameter of 6 mm. The conductors are separated by air. The inner conductor carries a constant current of 1.5 A flowing in the positive z direction and the outer conductor carries a constant current of 1.5 A flowing in the negative z direction.

- (a) Sketch the magnetic flux density, showing its direction. [5]
- (b) Calculate the magnetic flux density at radii of 1.5 mm, 2.5 mm and 4 mm. [5]

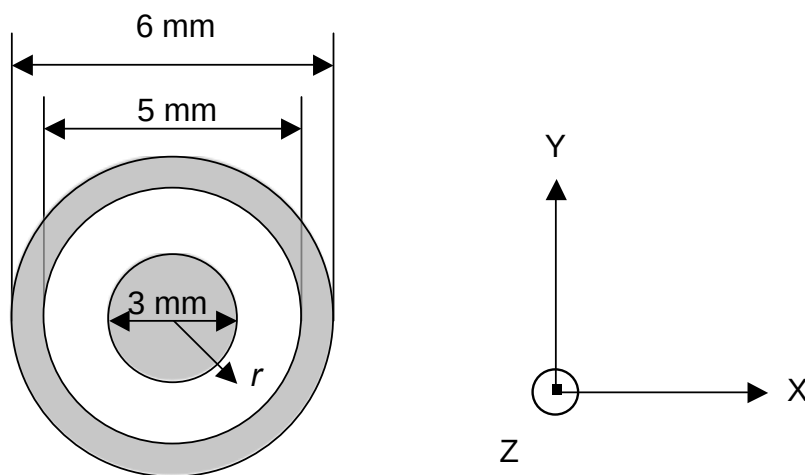


Fig. 8